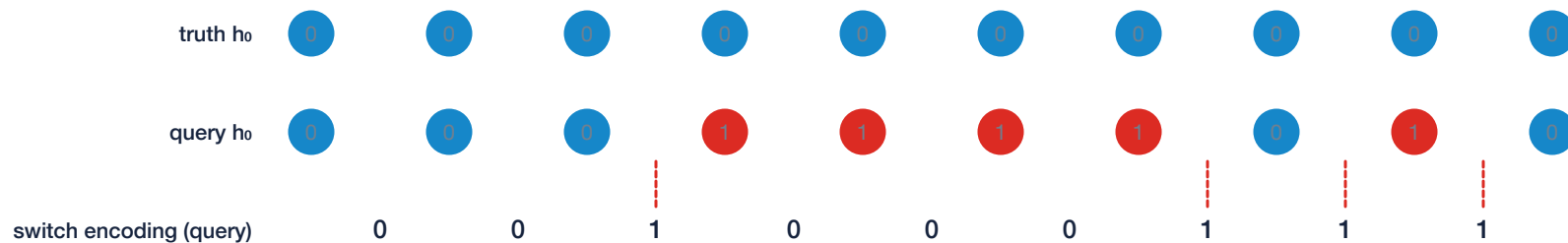


## a Switch errors, flips and Hamming distance within one block



switch errors = number of positions where the switch encodings differ = 4

switch/flip decomposition: two adjacent switches (positions 7–8) form one flip; isolated switches = 2, flips = 1

Hamming distance = min over the two orientations of misassigned variants =  $\min(5, 5) = 5$  of 10 (50%)

(per-block minimum, summed over blocks, divided by variants compared)

## b Block statistics

Blocks are intersected: only variants heterozygous in both files and phased in both are compared; singleton blocks are ignored.

Phased SNV (%) = truth heterozygous SNVs phased by the query / all truth heterozygous SNVs (same for indels, SVs, 5mC).

Block N50 = span of the block at which the cumulative span of query blocks, sorted by length, reaches 50% of the total.

Switch error rate (%) = switch errors / assessed variant pairs  $\times$  100. Cross-tool comparisons use SNVs only (--only-snvs).

Implemented in longphase compare (re-implementation of whatshap compare; identical values, parallel over chromosomes).